ESC131 / ESC231

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RAMAIAH
Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

MAKE UP / SUPPLEMENTARY SEMESTER EXAMINATIONS – DECEMBER 2023

Program	:	**B.E. – Common to all Programs**	**Semester**	:	**I / II**
Course Name	:	**Introduction to Civil Engineering**	**Max. Marks**	:	**100**
Course Code	:	**ESC 131 / ESC231**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.
- Assume missing data suitably.

UNIT - I

1. a) Describe the scopes of following fields of Civil Engineering: CO1 (08)
   (i) Structural Engineering (ii) Geotechnical Engineering.
   b) Elaborate on the properties of construction chemicals used. CO1 (06)
   c) Describe the following (i) Foundation and its types (ii) Lintel (iii) Chejja. CO1 (06)

2. a) Describe the scopes of following fields of Civil Engineering: CO1 (08)
   (i) Surveying (ii) Environmental Engineering.
   b) A builder has procured bricks for his building. Bricks has the following properties. Suggest their suitability for use in load bearing walls. They are red in colour, crushable at 5 N/mm^2, breaks when dropped from 0.5m on the ground, weight increased by 30% when soaked in water for a day. Give justification for your answer. CO1 (06)
   c) Differentiate between reinforced cement concrete and pre stressed cement concrete. CO1 (06)

UNIT - II

3. a) Enumerate and explain any four sources of air pollution in an urban area. CO2 (08)
   b) Explain in brief the process of solid waste management. CO2 (06)
   c) Describe the process of achieving sustainable development model. CO2 (06)

4. a) Brief out the various causes of flood in an urban area. Also explain about the relevant urban flood controls measures that can be adopted. CO2 (08)
   b) Explain in detail how rain water harvesting can be effectively implemented in an residential building. CO2 (06)
   c) With respect to transportation, brief out the implementation of smart city concept. CO2 (06)

UNIT - III

5. a) Draw relevant sketches to explain in detail the following support systems: CO3 (08)
   i) Simple Support
   ii) Hinged Support
   iii) Roller Support
   iv) Fixed Support.
   b) State and prove Varignons Theorem. CO3 (06)

	c)	Calculate the magnitude, direction and position of the resultant with respect to point O as shown in Fig. 5(c).	CO3	(06)

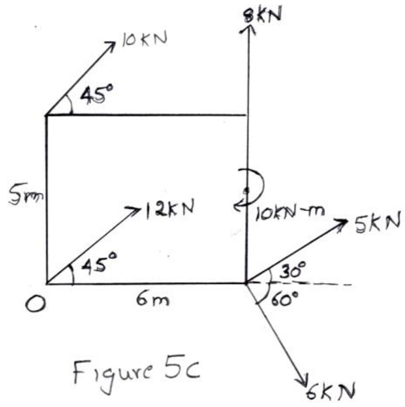


Figure 5c

6.	a)	Determine the support reactions provided by the compound beams shown in Fig 6(a).	CO3	(10)

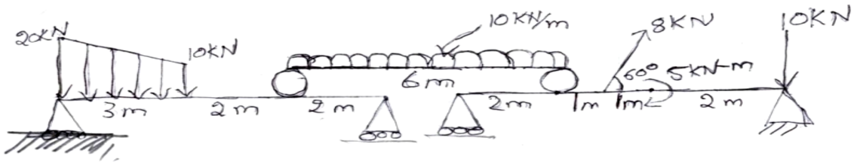


Figure 6 a

	b)	With relevant sketches explain in brief the different types of beams.	CO3	(05)
	c)	Determine the reactions at all the contact surfaces shown in the Fig. 6(c).	CO3	(05)

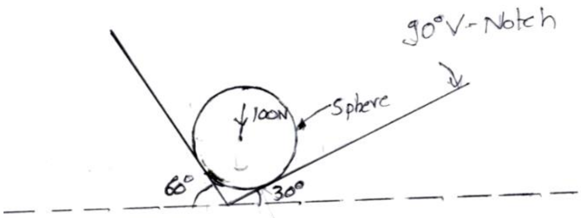
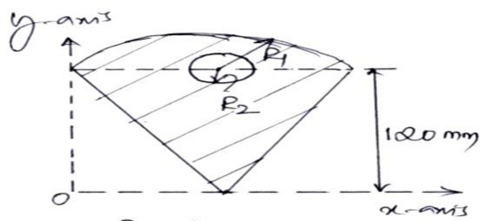


Figure 6c

UNIT- IV

7.	a)	Locate the Centroid of the shaded area with respect to the axes shown in Fig. 7(a).	CO4	(08)



$R_1 = 45$ mm
$R_2 = 20$ mm
Fig Q7 (a)

	b)	Derive the coordinates of the rectangular section from the first principles.	CO4	(06)
	c)	Define the following terms: (i) Coefficient of friction (ii) Angle of friction (iii) Limiting friction.	CO4	(06)

8. a) Calculate the value of W required in the Fig.8(a) CO4 (08)
   (i) To cause the body to move in the upward direction.
   (ii) To cause the body to move in the downward direction. Take μ=0.31.

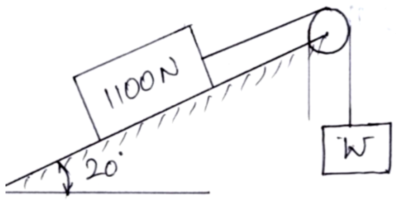


Fig. 8(a)

b) Determine the Centroid of the plane lamina shown in the Fig. 8(b) about the x axis. CO4 (06)

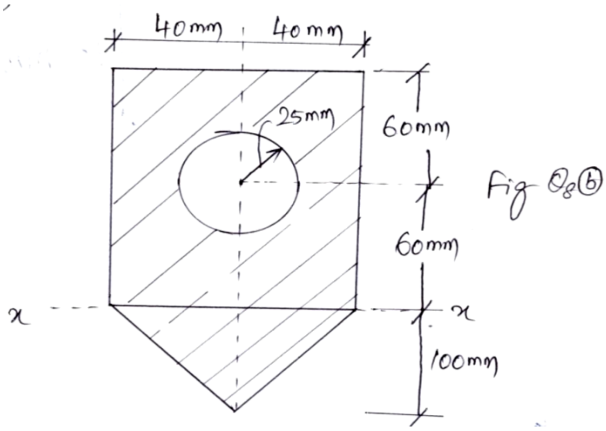


Fig Q_8(b)

c) A block weighing 1600N rests on a plane inclined at 25^0 to the horizontal as shown in Fig. 8(c). If μ=0.28, find the force required to push the block up the plane when the line of action of the force makes an angle of 25^0 with the plane. CO4 (06)

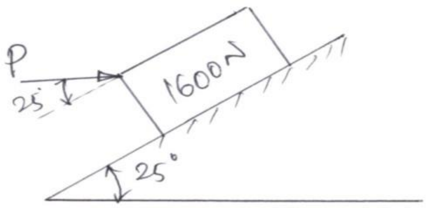


Fig Q_8(c)

UNIT- V

9. a) State and prove perpendicular axis theorem. CO5 (06)
   b) Derive an expression for moment of inertia of a triangle about its base. CO5 (06)
   c) Calculate the polar moment of inertia of the shaded area as shown in Fig.9(c). CO5 (08)

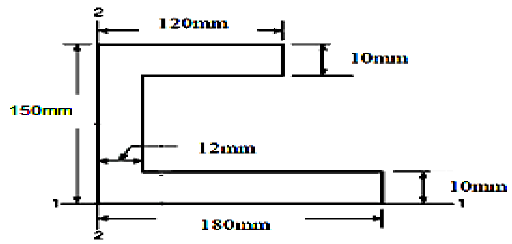


Fig.9(c)

10. a) State and prove Parallel axis theorem. CO5 (06)

b) Discuss the Polar moment of Inertia and the Radius of gyration. CO5 (06)

c) Determine the second moment of shaded area for the section shown in Fig.10(c) about AB. Take R_1 = 50 mm and R_2 = 20 mm. CO5 (08)

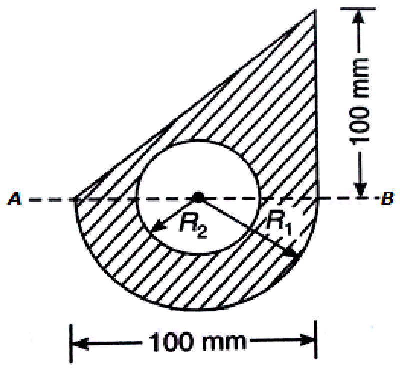


Fig. 10(c)

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